

## Unit 4 Polynomial Functions Study Guide

**Objectives...**Remember that big colored sheet of paper you taped into your notebook at the beginning of the unit??

### A. Polynomial Properties

- ☐ I can identify the degree and leading coefficient of a polynomial function.
- ☐ I can evaluate polynomials of different degrees.
- ☐ I can add, subtract, and multiply polynomials.
- ☐ I can factor polynomials using grouping, difference of squares, and sum/difference of cubes.
- ☐ I can determine the end behavior of polynomials.

### B. Dividing Polynomials

- ☐ I can divide polynomials using long division and synthetic division.
- ☐ I can use the Fundamental, Remainder, and Rational Root Theorems.

### C. Graphing Polynomials

- ☐ I can find x-intercepts of polynomials.
- ☐ I can determine the multiplicity of zeros of a polynomial.

- ☐ I can graph a polynomial function.
- ☐ I can solve applications involving polynomials.

### Essential Questions

1. What is a polynomial?
2. How do we use basic operations (add, subtract, multiply) with polynomials?
3. How do we factor polynomials?
4. What is end behavior?
5. How do we determine the end behavior of polynomials?
6. How do we divide polynomials?
7. What is the Fundamental Theorem?
8. What is the Remainder Theorem?
9. What is the Rational Root Theorem?
10. How do we graph a polynomial?
11. How do we find x-intercepts of polynomials?
12. What is multiplicity?
13. How do we find multiplicity of zeros?
14. How can we apply all of this to real applications?

### Topics Covered

**How can I practice/study for this assessment?**

*Your notes, practice, and homework creates a "practice guide"!*

1. Complete any unfinished practice problems
2. Redo practice problems
3. DeltaMath
4. Visit our Google Classroom page for digital versions of in class activities!!
5. Visit our Google Classroom page for any additional digital versions of practice!!

## Unit 4 Polynomial Functions Practice Guide

1. **CALCULATOR.** Priya wants to make an open-top box by cutting out corners of a 22 inch by 28 inch piece of poster board and then folding up the sides. The volume in cubic inches of the open-top box is a function of the side length in inches of the square cutouts.
  - a. Write an expression for  $V(x)$ .
  - b. What is a reasonable domain for  $V$  in this context?
  - c. When is the volume increasing? When is the volume decreasing?
  - d. What size cutout will maximize the volume? What is the maximum volume?
  
2. Simplify the polynomials using the given operation
  - a.  $(9x^5 + 5x^4 - 9x^2 + 10x) - (12x^5 + 2x^4 - x^2 - 9)$
  - b.  $(8x^7 - 6x^5 + 4x^3 - 6x) + (15x^6 + 4x^5 - 3x^3 + 2)$
  - c.  $(3x^2 - 6x - 5)(x^4 + 2x^2 + 5x)$
  - d. When  $2bx^4 + x^2 - b$  is added to  $3x^3 - dx^2 + cx + a$  the result is  $6x^4 + bx^3 - 4x^2 + 2x - 2$ . Find a, b, c, and d.
  
3. Factor the expressions completely.
 

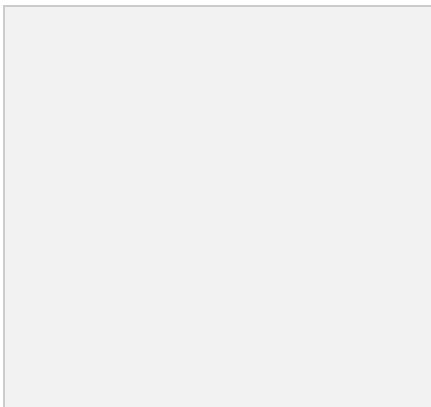
|                           |                             |
|---------------------------|-----------------------------|
| a. $2x^3 - 11x^2 + 12x$   | f. $2p^9 - 16p^6$           |
| b. $x^3 - 3x^2 + 5x - 15$ | g. $125 + 27x^3$            |
| c. $3x^4 - 432x^2$        | h. $125m^3 - 64$            |
| d. $a^3b - 8b$            | i. $54w^7 - 16w^4$          |
| e. $-10x^3 - 5x^2 + 5x$   | j. $25s^3 + 100s^2 - s - 4$ |
  
4. **CALCULATOR.** The profit  $P$  (in millions of dollars) for a DVD manufacturer can be modeled by  $P = -6x^3 + 72x$ , where  $x$  is the number (in millions) of DVDs produced. How many DVDs need to be produced for the company to yield a profit of \$96 million?
  
5. For each, identify the degree, leading coefficient, zeros and their multiplicity, y-intercept, and end behavior. Then, graph the function.
 

|                                     |                                      |
|-------------------------------------|--------------------------------------|
| a. $f(x) = (x - 1)^3(x + 1)$        | c. $h(x) = -(x + 3)^2(x - 1)(x - 5)$ |
| b. $g(x) = -5(x - 1)(x - 2)(x - 3)$ | d. $g(x) = 2(x + 2)^2(x + 4)^2$      |
  
6. Graph each polynomial. Be sure to find the following: y-intercept, roots/zeros (& the division to find these), multiplicity of each root/zero, end behavior (lead coefficient and degree).
 

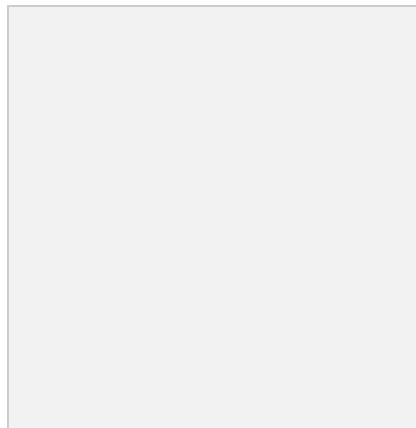
|                                                             |                                     |
|-------------------------------------------------------------|-------------------------------------|
| a. $f(x) = x^3 + 4x^2 - 41x + 36$ , known factor of $x - 4$ | g. $f(x) = x^4 + 6x^3 + 12x^2 + 8x$ |
| b. $f(x) = x^3 + 5x^2 - 4x - 20$                            | h. $m(x) = -2x^3 + 8x^2 + 8x - 32$  |
| c. $g(x) = -3x^3 - 9x^2 + 27x + 81$                         | i. $y = x^3 + 2x^2 - 15x$           |
| d. $k(x) = 4x^3 + 4x^2 - 81x - 81$                          | j. $y = -x^3 + 5x^2$                |
| e. $f(x) = x^4 + 3x^3 - 3x^2 - 11x - 6$                     | k. $f(x) = x^3 + 10x^2 + 31x + 30$  |
| f. $f(x) = x^3 + 7x^2 + 15x + 9$                            | l. $f(x) = 6x^3 - 11x^2 - 26x + 15$ |

7. For each graph below, identify the x-values for which (A) the function is increasing, (B) where the function is decreasing, (C)  $f(x) > 0$ , and (D)  $f(x) < 0$ . Then, (E) describe the end behavior, and (F) describe the degree and leading coefficient, (G) write an equation for the polynomial below.

a.



b.



8. Divide.

a.  $(4x^3 + 8x^2 + 13x + 5) \div (2x + 1)$

b.  $\frac{2x^3 + x^2 - 24x + 27}{2x - 3}$

c.  $(3x^3 + x^2 - 18x + 18) / (x + 3)$

d.  $\frac{x^4 + 6x^3 + 10x^2 + 29x + 3}{x^2 + x + 4}$

e.  $(3x^4 + 4x^3 - 7x^2 - 14x - 2) \div (x + 1)$

f.  $\frac{2x^3 + 3x^2 + 10x + 15}{2x + 3}$

g.  $\frac{4x^3 + 12x^2 + 7x + 23}{x + 3}$

9. Use the Remainder Theorem to answer the following problems.

a.  $f(x) = 5x^4 + 3$ , what is the remainder when divided by  $x - 1$ ?

b.  $\frac{2x^2 - x + m}{x - 1}$  has a remainder of 3. What is the value of m?

10. Write two different equations for a polynomial function whose graph intersects the horizontal axis at -7, 8, and -1/2.